

2024, Mar. 2024 Release LOC:Acute Adult

Pulmonary Embolism

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2, 3, 4)Episode Day 1 ^(1, 3, 4)Episode Day 2 ^(3, 4, 37)Episode Day 3-4 ^(4, 62)

Episode Day 5

Discharge Screens ⁽⁶⁶⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for the patient who is diagnosed with pulmonary embolism (PE) that has been confirmed by imaging.

Introduction:

PE is a form of venous thromboembolic (VTE) disease, a condition in which a clot forms within a vein, then dislodges and propagates into the branch(es) of the pulmonary circulation. These thrombi commonly form within the pelvic or lower extremity veins. Common clinical symptoms of PE include breathlessness, pleuritic chest pain, hemoptysis, tachycardia, and/or hypoxemia; however, in the setting of massive PE, a patient can present with hemodynamic compromise, including confusion, syncope, hypotension, shock, or sudden death (Khan et al., Lancet 2021, 398: 64-77).

Evaluation and Treatment:

Clinical findings alone are insufficient to establish a diagnosis of PE. The combination of symptoms and clinical findings, as well as the existence of VTE risk factors, can assist in determining which patients have a high probability of PE and who should have further diagnostic testing. Patients with a high clinical probability determined by clinical judgment or use of a clinical score (e.g., Wells score, Geneva score) should proceed to imaging to rule out PE; however, those with a low or moderate probability should have D-dimer testing. D-dimer testing has a high negative predictive value but a low positive predictive value, meaning that it can help exclude thrombosis if D-dimer is normal, but elevated D-dimer levels do not confirm the diagnosis. Low-risk patients with a positive D-dimer may require further imaging studies to confirm the diagnosis of PE (Khan et al., Lancet 2021, 398: 64-77; Konstantinides et al., Eur Heart J 2020, 41: 543-603). The first line of imaging for suspected PE is computed tomographic pulmonary angiography (CTPA), which has high specificity and sensitivity. When CTPA is contraindicated or unavailable, a ventilation-perfusion (V/Q) lung scan or magnetic resonance pulmonary angiography (MRPA) may be performed (Konstantinides et al., Eur Heart J 2020, 41: 543-603; American College of Radiology, ACR appropriateness criteria: suspected pulmonary embolism. 2016). Anticoagulation is the primary treatment for PE. However, in severe cases, active thrombus removal (e.g., catheter-directed thrombolysis) may be necessary. Categorizing the severity of a PE can help in determining which patients require more intensive monitoring and treatment. The American Heart Association (AHA) and the European Society of Cardiology (ESC) classify PE severity into three categories. High-risk (ESC) or massive (AHA) PE is characterized by hypotension or the need for vasopressor support. Intermediate risk (ESC) or submassive (AHA) PE is defined as right ventricular (RV) strain without hypotension. ESC also includes patients with a simplified Pulmonary Embolism Severity Index (sPESI) score of ≥ 1 in this category. Low-risk is defined as those who do not meet the other two categories (Giri et al., Circulation 2019, 140: e774-e801).

While the AHA and ESC risk stratification categories can assist clinicians in identifying high-risk patients, the sPESI can help identify low-risk PE patients who may be suitable for outpatient anticoagulation treatment or early discharge.

Patients with a sPESI score of 0 are considered low risk, while the presence of 1 or more risk factors significantly increases mortality to almost 11%(0.11). The sPESI includes the following clinical risk factors (Konstantinides et al., Eur Heart J 2020, 41: 543-603):

- Age > 80 years
- History of cancer
- History of heart failure or chronic lung disease
- Heart rate $\geq$ 110 beats per minute
- Systolic blood pressure < 100 mmHg
- Oxygen saturation < 90%(0.90)on room air

Other scoring systems, such as the Pulmonary Embolism Severity Index (PESI) and the Hestia criteria, may assist clinicians in determining the disposition of a patient. Patients who fall within PESI classes I or II or who meet the Hestia criteria can be considered for outpatient management of PE. The need to evaluate for RV dysfunction in patients selected for outpatient treatment remains a debate given that a meta-analysis showed the presence of RV dysfunction among patients classified as low-risk. Therefore, further testing with an echocardiogram and cardiac biomarkers should be considered in these patients (Yoo et al., Cochrane Database Syst Rev 2022, 5: Cd010019; Konstantinides et al., Eur Heart J 2020, 41: 543-603).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Stevens et al., Chest 2021, 160: e545-e608; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Ortel et al., Blood Adv 2020, 4: 4693-738).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾– Pulmonary embolism and anticoagulation (includes PO) ^(6, 7)● **ACUTE, One:** ⁽⁸⁾● Anticoagulation (includes PO) and, ≥ **One:** ^(7, 9)– Heart failure requiring IV diuretic therapy or titration of oral diuretic therapy ⁽¹⁰⁾– O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(11, 12, 13)● UFH, continuous and, ≥ **One:** ⁽¹⁴⁾– Antiphospholipid antibody syndrome ⁽¹⁵⁾– Bleeding, active or increased risk ⁽¹⁶⁾– BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁷⁾– Creatinine clearance ≤ 30 mL/min ⁽¹⁸⁾– HIT, suspected or by history and anticoagulation ⁽¹⁹⁾– IVC filter placement performed ≤ 24h ⁽²⁰⁾● Pretreatment or bridging anticoagulation, **All:** ⁽²¹⁾

– Patient or caregiver unable to administer

– Outpatient services unavailable ⁽²²⁾– Anticoagulation ⁽²³⁾● **INTERMEDIATE, One:** ⁽²⁴⁾● Right ventricular dysfunction, ≥ **One:** ⁽²⁵⁾● Suspected by biomarkers, ≥ **One:** ⁽²⁶⁾– BNP or NT-proBNP > ULN and > baseline ⁽¹³⁾– Troponin > ULN and > baseline ⁽¹³⁾– Confirmed by imaging ⁽²⁷⁾● **CRITICAL, ≥ One:** ⁽²⁸⁾● Thrombolytic therapy, ≥ **One:** ⁽²⁹⁾

– Catheter directed thrombolysis scheduled or performed ≤ 24h

– Systemic thrombolysis ≤ 24h

– ECMO or extracorporeal life support (ECLS) ⁽³⁰⁾– Hemodynamic monitoring, invasive ⁽³¹⁾– HFNC and respiratory monitoring every 1-2h ^(11, 32, 33)

– Mechanical ventilation

– NIPPV ⁽³⁴⁾– Oxygen ≥ 40%(0.40) ⁽¹¹⁾

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ^(1, 3, 4)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁵⁾– Pulmonary embolism and anticoagulation (includes PO) ^(6, 7, 35)● **ACUTE, One:** ⁽⁸⁾● Anticoagulation (includes PO) and, ≥ **One:** ^(7, 9)– Heart failure requiring IV diuretic therapy or titration of oral diuretic therapy ⁽¹⁰⁾– O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(11, 12, 13)● UFH, continuous and, ≥ **One:** ⁽¹⁴⁾– Antiphospholipid antibody syndrome ⁽¹⁵⁾– Bleeding, active or increased risk ⁽¹⁶⁾– BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁷⁾– Creatinine clearance ≤ 30 mL/min ⁽¹⁸⁾– HIT, suspected or by history and anticoagulation ⁽¹⁹⁾

- Hospital acquired and initiation of anticoagulation (includes PO) $\leq 24\text{h}$ ^(9, 36)
- IVC filter placement performed $\leq 24\text{h}$ ⁽²⁰⁾
- Pretreatment or bridging anticoagulation, **All:** ⁽²¹⁾
 - Patient or caregiver unable to administer
 - Outpatient services unavailable ⁽²²⁾
 - Anticoagulation ⁽²³⁾
- **INTERMEDIATE, One:** ⁽²⁴⁾
 - Right ventricular dysfunction, $\geq$ **One:** ⁽²⁵⁾
 - Suspected by biomarkers, $\geq$ **One:** ⁽²⁶⁾
 - BNP or NT-proBNP $>$ ULN and $>$ baseline ⁽¹³⁾
 - Troponin $>$ ULN and $>$ baseline ⁽¹³⁾
 - Confirmed by imaging ⁽²⁷⁾
- **CRITICAL, $\geq$ One:** ⁽²⁸⁾
 - Surgical embolectomy (**use General Surgical subset**)
 - Thrombolytic therapy, $\geq$ **One:** ⁽²⁹⁾
 - Catheter directed thrombolysis scheduled or performed $\leq 24\text{h}$
 - Systemic thrombolysis $\leq 24\text{h}$
 - ECMO or extracorporeal life support (ECLS) ⁽³⁰⁾
 - Hemodynamic monitoring, invasive ⁽³¹⁾
 - HFNC and respiratory monitoring every 1-2h ^(11, 32, 33)
 - Mechanical ventilation
 - NIPPV ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40)$ ⁽¹¹⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ^(3, 4, 37)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽³⁸⁾
 - Anticoagulation regimen established ⁽³⁹⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h of Observation for this condition, **One:**
 - For pulmonary embolism use Episode Day 2 at a higher level of care
 - For a condition other than pulmonary embolism use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁸⁾
 - Anticoagulation regimen, **One:**
 - Established ⁽³⁹⁾
 - Contraindicated ⁽⁴⁰⁾
 - O₂ sat within acceptable limits or home oxygen arranged ⁽⁴¹⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - ACS suspected (**use Acute Coronary Syndrome subset**)
 - Anticoagulation (includes PO) and, $\geq$ **One:** ^(7, 9)
 - O₂ sat $\leq 91\%(0.91)$ and $<$ baseline requiring supplemental oxygen ^(11, 12, 13)
 - Biomarkers, $\geq$ **One:** ^(43, 44)
 - BNP or NT-proBNP $>$ ULN and $>$ baseline ⁽¹³⁾
 - Troponin $>$ ULN and $>$ baseline ⁽¹³⁾
 - UFH, continuous and, $\geq$ **One:** ⁽¹⁴⁾
 - Antiphospholipid antibody syndrome ⁽¹⁵⁾
 - Bleeding, active or increased risk ⁽¹⁶⁾
 - BMI $\geq 40\text{ kg/m}^2$ or weight $> 120\text{ kg}$ ⁽¹⁷⁾
 - Creatinine clearance $\leq 30\text{ mL/min}$ ⁽¹⁸⁾

- HIT, suspected or by history and anticoagulation ⁽¹⁹⁾
- IVC filter placement performed $\leq 24\text{h}$ ⁽²⁰⁾
- Pain consistent with pulmonary embolism requiring analgesic $\geq 3\text{x}/24\text{h}$, $\leq 2\text{d}$ ⁽⁴⁵⁾
- Pretreatment or bridging anticoagulation, **All:** ⁽²¹⁾
 - Patient or caregiver unable to administer
 - Outpatient services unavailable ⁽²²⁾
- Intervention, **One:**
 - INR non-therapeutic and anticoagulation (includes PO) ⁽⁴⁶⁾
 - Anticoagulation $\leq 5\text{d}$ since initiation ^(47, 48)
- Bleeding, actual or suspected and, $\geq$ **One:** ^(49, 50)
 - Blood product transfusion
 - Hct or Hb monitoring at least $2\text{x}/24\text{h}$, $\leq 2\text{d}$
- COPD and, **Both:** ⁽⁵¹⁾
 - Persistent dyspnea and, $\geq$ **One:**
 - Arterial $\text{Po}_2 < \text{baseline}$ ^(13, 52)
 - Arterial or venous $\text{Pco}_2 > \text{baseline}$ ^(13, 52)
 - $\text{O}_2 \text{ sat} \leq 91\%(0.91)$ and $< \text{baseline}$ ⁽¹³⁾
 - Increased work of breathing, $\geq$ **One:**
 - Difficulty taking PO
 - Prefers sitting ⁽⁵³⁾
 - Talks in phrases
 - Use of accessory muscles
- Intervention, **Both:**
 - Bronchodilator $\geq 4\text{x}/24\text{h}$ ⁽⁵⁴⁾
 - Corticosteroid (includes inhaled) ⁽⁵⁵⁾
- Heart failure and, **Both:** ⁽⁵⁶⁾
 - Dyspnea $> \text{baseline}$ and, $\geq$ **One:** ⁽¹³⁾
 - $\text{O}_2 \text{ sat} \leq 91\%(0.91)$ and $< \text{baseline}$ ⁽¹³⁾
 - Worsening peripheral edema
 - Heart failure confirmed by imaging ⁽⁵⁷⁾
 - Intervention, $\geq$ **One:** ⁽⁵⁸⁾
 - Diuretic, **One:** ⁽⁵⁹⁾
 - $\geq 2\text{x}/24\text{h}$ (includes PO)
 - At least $1\text{x}/24\text{h}$ and requires daily medication adjustment (includes PO) and, $\geq$ **One:** ⁽⁶⁰⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹³⁾
 - Creatinine increased from baseline ⁽¹³⁾
 - Dialysis
 - Post Critical care $\leq 24\text{h}$
- INTERMEDIATE, **One:**
 - Right ventricular dysfunction confirmed by echocardiogram $\leq 3\text{d}$ ⁽⁶¹⁾
- CRITICAL, $\geq$ **One:**
 - Thrombolytic therapy, $\geq$ **One:** ⁽²⁹⁾
 - Catheter directed thrombolysis scheduled or performed $\leq 24\text{h}$
 - Systemic thrombolysis $\leq 24\text{h}$
 - Post thrombolytic therapy $\leq 24\text{h}$
 - ECMO or extracorporeal life support (ECLS) ⁽³⁰⁾
 - Hemodynamic monitoring, invasive ⁽³¹⁾
 - HFNC and respiratory monitoring every $1-2\text{h}$ ^(11, 32, 33)
 - Mechanical ventilation
 - NIPPV ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40)$ ⁽¹¹⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 3-4, **One:** ^(4, 62)

● ACUTE, One:

● Responder, discharge expected today if clinically stable last 12h, All: ⁽³⁸⁾

● Anticoagulation regimen, One:

- Established ⁽³⁹⁾
- Contraindicated ⁽⁴⁰⁾
- Laboratory values within acceptable limits ⁽⁴¹⁾
- O₂ sat within acceptable limits or home oxygen arranged ⁽⁴¹⁾

● Complication or comorbidity, One:

- No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
- Complication or comorbidity and clinically stable for discharge, ≥ One:
 - Bleeding controlled or resolved
 - COPD exacerbation resolving
 - Heart failure resolving
 - Pain controlled or manageable

● Partial responder, not clinically stable for discharge and requires continued stay, ≥ One:

● UFH, continuous and, ≥ One: ⁽¹⁴⁾

- Antiphospholipid antibody syndrome ⁽¹⁵⁾
- Bleeding, active or increased risk ⁽¹⁶⁾
- BMI ≥ 40 kg/m² or weight > 120 kg ⁽¹⁷⁾
- Creatinine clearance ≤ 30 mL/min ⁽¹⁸⁾
- Pretreatment or bridging ⁽⁶³⁾
- O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(11, 12, 13)
- HIT and anticoagulation ^(19, 64)
- IVC filter placement performed ≤ 24h ⁽²⁰⁾

● Pain consistent with pulmonary embolism and, One: ⁽⁴⁵⁾

- Analgesic ≥ 3x/24h, ≤ 2d
- Transition to PO analgesic ≤ 24h

● Pretreatment or bridging anticoagulation, All: ⁽²¹⁾

- Patient or caregiver unable to administer
- Outpatient services unavailable ⁽²²⁾

● Intervention, One:

- INR non-therapeutic and anticoagulation (includes PO) ⁽⁴⁶⁾
- Anticoagulation ≤ 5d since initiation ^(47, 48)

● Bleeding, actual or suspected and, ≥ One: ^(49, 50)

- Blood product transfusion
- Hct or Hb monitoring at least 2x/24h, ≤ 2d

● COPD and, Both: ⁽⁵¹⁾

● Persistent dyspnea and, ≥ One:

- Arterial Po₂ < baseline ^(13, 52)
- Arterial or venous Pco₂ > baseline ^(13, 52)
- O₂ sat ≤ 91%(0.91) and < baseline ⁽¹³⁾

● Increased work of breathing, ≥ One:

- Difficulty taking PO
- Prefers sitting ⁽⁵³⁾
- Talks in phrases
- Use of accessory muscles

● Intervention, Both:

- Bronchodilator ≥ 4x/24h ⁽⁵⁴⁾
- Corticosteroid (includes inhaled) ⁽⁵⁵⁾

● Heart failure and, Both: ⁽⁵⁶⁾● Dyspnea > baseline and, ≥ One: ⁽¹³⁾

- O₂ sat ≤ 91%(0.91) and < baseline ⁽¹³⁾
- Worsening peripheral edema

- Heart failure confirmed by imaging ⁽⁵⁷⁾
- **Intervention, ≥ One:** ⁽⁵⁸⁾
 - **Diuretic, One:** ⁽⁵⁹⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** ⁽⁶⁰⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹³⁾
 - Creatinine increased from baseline ⁽¹³⁾
 - Dialysis
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Right ventricular dysfunction confirmed by echocardiogram ≤ 3d ⁽⁶¹⁾
 - HFNC and respiratory monitoring every 3-4h ^(11, 32, 33)
 - NIPPV ⁽³⁴⁾
 - Oxygen ≥ 40%(0.40) ⁽¹¹⁾
- **CRITICAL, ≥ One:**
 - Post thrombolytic therapy ≤ 24h
 - ECMO or extracorporeal life support (ECLS) ⁽³⁰⁾
 - Hemodynamic monitoring, invasive ⁽³¹⁾
 - HFNC and respiratory monitoring every 1-2h ^(11, 32, 33)
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽³⁴⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁶⁵⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁸⁾
 - **Anticoagulation regimen, One:**
 - Established ⁽³⁹⁾
 - Contraindicated ⁽⁴⁰⁾
 - Laboratory values within acceptable limits ⁽⁴¹⁾
 - O₂ sat within acceptable limits or home oxygen arranged ⁽⁴¹⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁴²⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Bleeding controlled or resolved
 - COPD exacerbation resolving
 - Heart failure resolving
 - Pain controlled or manageable
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁶⁶⁾**● HOME, Both:****● Level of care appropriateness, Both:**

- Home environment safe and accessible ⁽⁶⁷⁾
- Patient or caregiver demonstrates ability to manage care

● Pulmonary embolism discharge planning, All:

- Provide comprehensive written discharge and teaching instructions ⁽⁶⁸⁾

● Education on the importance of, All:

- Anticoagulation management ⁽⁶⁹⁾
- Caution when using nonsteroidal anti-inflammatory medications or aspirin ⁽⁷⁰⁾
- Pulmonary embolism prevention methods ⁽⁷¹⁾

- Medication reconciliation ⁽⁷²⁾

- Follow-up care planned ⁽⁷³⁾

- Identify and address transportation needs

● HOME CARE, Both:**● Level of care appropriateness, All:**

- Home environment safe and accessible ⁽⁶⁷⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(74, 75)

● Skilled services, ≥ One: ⁽⁷⁶⁾

- Skilled nursing ⁽⁷⁷⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁷⁸⁾

● Pulmonary embolism discharge planning, All:

- Provide comprehensive written discharge and teaching instructions ⁽⁶⁸⁾

● Education on the importance of, Both:

- Anticoagulation management ⁽⁶⁹⁾
- Pulmonary embolism prevention methods ⁽⁷¹⁾

- Medication reconciliation ⁽⁷²⁾

- Follow-up care planned ⁽⁷³⁾

- Identify and address transportation needs

● BEHAVIORAL HEALTH, Both:**● Level of care appropriateness, One:**

- Support systems available ⁽⁷⁹⁾

● Skilled treatment, ≥ One:

- Clinical assessment ⁽⁸⁰⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽⁸¹⁾

● SKILLED MEDICAL OR THERAPY, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ One:

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁷⁷⁾

● Therapy, All:

- Goal directed therapy with rehabilitation potential ^(82, 83)
- Functional impairments requiring at least supervision ⁽⁸⁴⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁷²⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner

- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁷⁷⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(82, 83)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁴⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷²⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(82, 83)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁴⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷²⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(82, 83)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸⁴⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁸⁵⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷²⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● Level of care appropriateness, **Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁸⁶⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(87, 88)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁸⁹⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷²⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
175 PULMONARY EMBOLISM WITH MCC OR ACUTE COR PULMONALE	n/a	4	CMS GMLOS
176 PULMONARY EMBOLISM WITHOUT MCC	n/a	2.5	CMS GMLOS
PULMONARY EMBOLISM	19%	3.6	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List
- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress after 24h

- Anticoagulation initiated
- Dyspnea decreasing
- O₂ saturation improving
- Pain improving or controlled
- Vital signs stabilizing

Consider:

- Anticoagulation teaching
- Decreasing O₂
- Discharge planning

4:

Prolonged oxygen therapy expected, arrange home oxygen:

- Home oxygen therapy should be considered in a clinically stable patient who continues to require supplemental oxygen when there has been no change in oxygen requirements within last 2 days (**Home**)

Potential risk for readmission (**Home Care, Readmission Reduction Program**)

- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission Reduction Program**). Provide patient education including: anticoagulation medication regimen training, nutritional teaching for patients prescribed warfarin, and over-the-counter medications to avoid (e.g., nonsteroidal anti-inflammatory drugs, aspirin).
- Subtherapeutic anticoagulation and arrange anticoagulation monitoring (**Home Care**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Anticoagulation may include the following:

- Direct oral anticoagulants (DOACs)
- Low molecular weight heparin (LMWH)
- Fondaparinux

7:

Initial anticoagulation with low molecular weight heparin (LMWH) or fondaparinux is preferable over unfractionated heparin (UFH) because they have a lower risk of causing severe bleeding and heparin-induced thrombocytopenia. In addition, LMWH and fondaparinux do not require monitoring of therapeutic levels on a regular basis. The use of UFH is currently limited to individuals with overt hemodynamic instability or imminent hemodynamic decompensation who may require primary reperfusion therapy, and those with comorbidities that preclude the use of other anticoagulants. Direct oral anticoagulants (DOAC) can also achieve rapid anticoagulant effects; therefore, they can be used for anticoagulation treatment for acute PE. The American Society of Hematology (ASH) suggests using DOACs over vitamin K antagonists (VKAs), except in the setting of severe reduction in kidney function, moderate to severe liver disease (e.g., Child-Pugh class B or C), or antiphospholipid syndrome (Konstantinides et al., Eur Heart J 2020, 41: 543-603; Ortel et al., Blood Adv 2020, 4: 4693-738).

8:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

9:

Anticoagulation may include the following:

- Direct oral anticoagulants (DOACs)
- Low molecular weight heparin (LMWH)
- Unfractionated heparin (UFH), continuous
- Fondaparinux

10:

Chronic heart failure in the setting of acute pulmonary embolism (PE) has been associated with a longer length of stay, an increase in mortality, acute respiratory distress, cardiac arrest, cardiogenic shock, major bleeding, ventricular arrhythmias, and the need for extracorporeal membrane oxygenation (ECMO) and ventilatory assistance compared to those without chronic heart failure (Quintero-Martinez et al., Am J Cardiol 2023, 195: 17-22).

11:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

12:

Hypoxemia is caused by a mismatch between ventilation and perfusion in patients with pulmonary embolism (PE) and is an indicator of severe disease (Konstantinides et al., Eur Heart J 2020, 41: 543-603).

13:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

14:

Unfractionated heparin (UFH) is preferable over low molecular weight heparin (LMWH) for the treatment of venous thromboembolism (VTE) in the following conditions (Mazzolai et al., Eur Heart J 2018, 39: 4208-18):

- Antiphospholipid antibody syndrome
- Bleeding or risk for bleeding
- BMI ≥ 40 kg/m² or weight > 120 kg
- Renal failure (CrCl < 30 mL/min)

15:

Thrombotic events are common in patients with antiphospholipid antibody syndrome. The ease of use and lack of need for monitoring have led to interest in using direct oral anticoagulants (DOACs) for management, but warfarin continues to be preferred by many providers. While the use of low molecular weight heparin (LMWH) in bridging to warfarin may be appropriate for some patients, unfractionated heparin is preferred in patients with known or suspected catastrophic antiphospholipid syndrome, or in patients with anticoagulant-refractory disease who may require higher intensity dosing (and as such, would incur an increased risk of bleeding) (Cohen et al., J Thromb Haemost 2021, 19: 892-908).

16:

The risk for major bleeding is highest during the first month of anticoagulant therapy, making it imperative that a patient's risk for bleeding be assessed at the time of initiation of anticoagulant treatment (Konstantinides et al., Eur Heart J 2020, 41: 543-603). In patients with an increased risk for bleeding, unfractionated heparin (UFH) may be the preferred initial treatment due to its short half-life and reversibility with intravenous (IV) protamine sulfate (Mazzolai et al., Eur Heart J 2018, 39: 4208-18). Although there are several validated venous thromboembolism (VTE) bleeding risk assessment tools, studies have yet to be conducted to determine their impact on treatment decisions; therefore, the International Society of Thrombosis and Haemostasis (ISTH) did not make a recommendation on which bleeding risk score to use prior to starting anticoagulation therapy (den Exter et al., J Thromb Haemost 2022, 20: 1910-9). As a result, the bleeding risk factors for this criterion were derived from a combination of the best-studied risk assessment tools for major bleeding in patients with VTE, guidelines, and literature focusing on a moderate to strong association with the risk of bleeding (den Exter et al., J Thromb Haemost

2022, 20: 1910-9; Klok and Huisman, Blood 2020, 135: 724-34; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52).

Instruction:

A patient who has a scheduled or anticipated procedure or surgery and/or has one or more of the following risk factors meets for this criterion:

- Major bleeding within the last 3 months (e.g., bleeding that caused a fall in hemoglobin ≥ 2 g/dl, blood transfusion of ≥ 2 units, or symptomatic bleeding in a critical area or organ (e.g., intracranial, spinal, ocular, pericardia, articular, intramuscular with compartment syndrome, retroperitoneal)) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52)
- Renal failure (e.g., creatinine clearance 30-60 ml/min, receiving dialysis) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Active cancer (e.g., receiving treatment) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Klok et al., Thromb Haemost 2017, 117: 1164-70; Konstantinides et al., J Am Coll Cardiol 2016, 67: 976-90; Kearon et al., Chest 2008, 133: 454S-545S; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Anemia (e.g., hemoglobin < 13 g/dL in male sex assigned at birth, < 12 g/dL in female sex assigned at birth) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Klok et al., Thromb Haemost 2017, 117: 1164-70; Kearon et al., Chest 2016, 149: 315-52; Ruiz-Giménez et al., Thromb Haemost 2008, 100: 26-31)
- Age > 75 years (Konstantinides et al., Eur Heart J 2020, 41: 543-603)
- Concurrent use of platelet inhibitors or nonsteroidal anti-inflammatory drugs (NSAIDs) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Uncontrolled hypertension (e.g., systolic blood pressure ≥ 140 mmHg) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Brown et al., J Am Heart Assoc 2018, 7: epub; Klok et al., Thromb Haemost 2017, 117: 1164-70)
- Thrombocytopenia (e.g., platelets $< 100,000$ cu.mm(100×10^9 /L)) (den Exter et al., J Thromb Haemost 2022, 20: 1910-9; Kearon et al., Chest 2016, 149: 315-52)
- Liver failure (Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Previous stroke (hemorrhagic or ischemic), timeframe not defined by literature (Konstantinides et al., Eur Heart J 2020, 41: 543-603; Brown et al., J Am Heart Assoc 2018, 7: epub; Kearon et al., Chest 2016, 149: 315-52)
- Recent surgery (e.g., within the last 3 months) (Kearon et al., Chest 2016, 149: 315-52)

17:

Unfractionated heparin is often used in the management of acute thromboembolic events in patients with a body mass index (BMI) greater than or equal to 40 kg/m^2 or weight greater than 120 kg; this practice is based on mixed evidence for direct oral anticoagulants (DOACs) in this population, and concerns about low molecular weight heparin (LMWH) bioavailability. The 2016 International Society on Thrombosis and Haemostasis (ISTH) guidelines recommended against the use of DOACs in patients with a BMI greater than or equal to 40 kg/m^2 or weight greater than 120 kg, and a 2019 systematic review (19 studies, including pharmacokinetic, studies, phase III RCTs and post hoc analyses, and retrospective cohort studies) continued to suggest caution (Kido et al., Pharmacotherapy 2020, 40: 72-83); however, a recent meta-analysis (5 observational studies, $n=6585$) concluded that these agents were non-inferior to warfarin (Elshafei et al., J Thromb Thrombolysis 2021, 51: 388-96). Absorption of LMWH administered subcutaneously may be variable in patients with obesity, and treatment with intravenous unfractionated heparin provides a reliable means of anticoagulation for patients with acute thromboembolic events (Lee et al., Clin Drug Investig 2020, 40: 33-40).

18:

Patients with creatinine clearance less than or equal to 30 mL/min should be treated with unfractionated heparin (UFH) when parenteral anticoagulation is necessary. Low molecular weight heparin (LMWH) and certain direct oral anticoagulants are predominantly excreted by the kidneys, and therefore may accumulate in patients with renal insufficiency causing an increased risk of bleeding (Kakkos et al., Eur J Vasc Endovasc Surg 2021, 61: 9-82).

19:

Heparin-induced thrombocytopenia (HIT) results from sensitization to the heparin molecule. Usually, it occurs between 5 to 10 days after the current heparin exposure but may occur as early as minutes to hours after exposure. HIT is a greater than 50%(0.50) serial reduction in a patient's baseline platelet count. A direct thrombin inhibitor (DTI) (e.g., argatroban, bivalirudin) is the initial treatment for patients with active HIT, a history of HIT, or those with a strong clinical suspicion of HIT who require anticoagulation. Although the American Society of Hematology (ASH) guidelines suggest the use of fondaparinux or direct oral anticoagulants (DOACs), the recommendation is conditional and based on very low certainty of evidence. In addition, these agents are considered off-label use in the treatment of HIT since they have yet to be approved by the U.S. Food and Drug Administration (FDA) for this condition. For patients who require long-term anticoagulation, warfarin should not be initiated until platelet count recovers (e.g., platelet count greater than $150 \times 10^9/L$). Additionally, a 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92; Guyatt et al., Chest 2012, 141: 48S-52S).

20:

Although routine use of inferior vena cava (IVC) filters in the treatment of patients with venous thromboembolism (VTE) is not recommended, guidelines suggest IVC filter placement in the setting of acute pulmonary embolism (PE) in patients with a contraindication to anticoagulation therapy to potentially reduce the mortality from acute PE and short-term PE recurrence (Kaufman et al., J Vasc Interv Radiol 2020, 31: 1529-44).

21:

The American Society of Hematology (ASH) guidelines for the management of venous thromboembolism (VTE) recommend offering home treatment over hospital treatment for patients with pulmonary embolism (PE) with low risk for complications, determined by a clinical prediction score such as the Pulmonary Embolism Severity Index (PESI) or simplified Pulmonary Embolism Severity Index (sPESI) in addition to clinical judgment. However, this recommendation does not apply to individuals who have inadequate home support, a lack of access to medication, or a history of poor compliance (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients with these risk factors may require inpatient admission while receiving parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin) during warfarin bridging therapy or pre-treatment therapy for dabigatran or edoxaban.

22:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting
- Skilled nursing services unavailable (e.g., Home Care)

23:

Anticoagulation includes low molecular weight heparin (LMWH) and fondaparinux.

24:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

25:

Patients with acute pulmonary embolism (PE) and right ventricular dysfunction demonstrated by imaging (computed tomography pulmonary angiography and/or echocardiogram) or an increase in cardiac biomarkers are at increased risk for mortality and adverse clinical outcomes, and therefore should be monitored closely for hemodynamic instability (Konstantinides et al., Eur Heart J 2020, 41: 543-603).

26:

Elevated cardiac markers in patients with pulmonary embolism indicate an increased risk of myocardial injury due to the imbalance between demand and supply of oxygen. Patients require monitoring until echocardiography is performed to determine whether right ventricular dysfunction is present (Konstantinides et al., Eur Heart J 2020, 41: 543-603).

27:

Right ventricular dysfunction may be confirmed by:

- Echocardiogram
- Computed tomography pulmonary angiography (CTPA)

28:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

29:

Systemic thrombolytic therapy in patients with pulmonary embolism (PE) who are normotensive (e.g., systolic blood pressure $\geq$ 90 mmHg) has been shown to prevent cardiovascular collapse; however, it has been associated with increased major bleeding, including extracranial and intracranial hemorrhage. Additionally, rescue systemic thrombolytic therapy (e.g., administered after patients become hemodynamically unstable) also appears to be beneficial. As a result, the American College of Chest Physicians (ACCP) suggests that patients with a PE at the severe end of the spectrum without hypotension be treated with intensive anticoagulation, vigilant monitoring, and other supportive measures rather than systemic thrombolytic treatment unless hypotension is present (Stevens et al., Chest 2021, 160: e545-e608). However, if there are signs of clinical deterioration (e.g., progressive increase in heart rate, worsening gas exchange, signs of shock) and the bleeding risk is acceptable, systemic thrombolytic therapy is recommended even if the patient remains normotensive.

Evidence comparing catheter-directed therapy (CDT) to systemic thrombolysis is lacking; however, there is a higher certainty of evidence on the benefits of systemic thrombolysis compared to anticoagulation alone. As a result, the ACCP suggests the use of systemic thrombolysis over CDT, except when the patient has a high risk for bleeding, fails systemic thrombolysis, or develops shock that is likely to cause death before systemic thrombolysis can take effect, in which case CDT is recommended (Stevens et al., Chest 2021, 160: e545-e608).

30:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

31:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line,

pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

32:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

33:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

34:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

35:

Instruction: Patients requiring an unfractionated heparin continuous infusion should be reviewed at the Acute level of care.

36:

Hospital acquired refers to a condition that developed during the patient's current hospitalization. Criteria for hospital acquired conditions cannot be applied to patients who either have been discharged to home or transferred from a post-acute facility.

37:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

38:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

39:

An outpatient anticoagulation regimen may include low molecular weight heparin (LMWH), fondaparinux, warfarin, or a direct oral anticoagulant (DOAC). For patients who will be treated with warfarin, initiation of treatment must overlap with parenteral anticoagulation until a therapeutic international normalized ratio (INR) is achieved; therefore, some patients could be discharged on parenteral anticoagulant along with warfarin if discharged prior to completion of overlap therapy. Dabigatran and edoxaban, both DOACs, require 5 to 10 days of pretreatment with parenteral anticoagulation; therefore, these patients may be discharged on a parenteral anticoagulation with the plan to switch to dabigatran or edoxaban after pretreatment is complete (Ortel et al., Blood Adv 2020, 4: 4693-738). This criterion also includes patients with confirmed or a history of heparin-induced thrombocytopenia (HIT) who have achieved therapeutic anticoagulation and whose outpatient regimen has been established.

40:

Although anticoagulation is the recommended treatment for venous thromboembolism (VTE) (e.g., deep vein thrombosis (DVT), pulmonary embolism (PE)), there are some circumstances in which anticoagulation may be contraindicated (e.g., active bleeding). If a patient cannot be treated with anticoagulation, a retrievable inferior vena cava (IVC) filter may be inserted, and the patient could be discharged without anticoagulation (Ortel et al., Blood Adv 2020, 4: 4693-738). Patients who experience an episode of major bleeding while on anticoagulation therapy may also be discharged without anticoagulation treatment initially, since guidelines recommend resumption of oral anticoagulation within 90 days (Witt et al., Blood Adv 2018, 2: 3257-91).

41:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

42:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

43:

Abnormal cardiac biomarkers demonstrate myocardial injury. Patients with pulmonary embolism who have negative imaging for right ventricular dysfunction but have increased cardiac biomarkers require close monitoring for signs of cardiac deterioration (Kearon et al., Chest 2016, 149: 315-52).

44:

Instruction: Patient demonstrating right ventricular dysfunction on imaging should be reviewed at the Intermediate level of care.

45:

Chest pain is a common symptom of pulmonary embolism (PE) and is typically caused by pleural irritation from a

distal emboli resulting in pulmonary infarction (Konstantinides et al., Eur Heart J 2020, 41: 543-603). According to guidelines, patients with PE who require intravenous analgesics may benefit from initial treatment in the hospital (Ortel et al., Blood Adv 2020, 4: 4693-738).

InterQual® external peer reviewers agree that patients who experience pulmonary embolism and pleuritic chest or back pain, requiring IV analgesics, warrant further intervention and assessment in a hospital environment for worsening condition.

46:

Instruction: Patients who have not achieved a therapeutic INR on 2 consecutive days and continue on LMWH or fondaparinux will meet this criterion.

According to the American Society of Hematology (ASH), when warfarin is being initiated, it should be overlapped with parenteral anticoagulation (e.g., fondaparinux, low molecular weight heparin (LMWH)) until a therapeutic international normalized ratio (INR) of 2.0-3.0 is achieved (Ortel et al., Blood Adv 2020, 4: 4693-738).

47:

Patients to be treated with dabigatran or edoxaban, both direct oral anticoagulants (DOACs), require at least 5 days of parenteral anticoagulation prior to initiation (Ortel et al., Blood Adv 2020, 4: 4693-738).

Instruction: This criteria point is intended for patients who require continued hospitalization for pretreatment with parenteral anticoagulation (e.g., low molecular weight heparin (LMWH) or fondaparinux) prior to starting dabigatran or edoxaban.

48:

Anticoagulation includes low molecular weight heparin (LMWH) or fondaparinux.

49:

Patients who develop bleeding while receiving anticoagulation should be evaluated for an underlying source of bleeding. Evaluation may include upper and lower gastrointestinal (GI) studies to rule out underlying GI pathology or an imaging study to rule out intracranial bleeding (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3).

50:

Findings associated with bleeding may include (Janz, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 9 edn. 2018:1463-79.e3):

- Heart rate 100-120/min, sustained
- Hct or Hb decreased within last 24h
- Orthostatic hypotension
- Mental status change
- Presyncope or syncope

51:

A severe chronic obstructive pulmonary disease (COPD) exacerbation may be indicated by the following signs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Celli et al., Eur Respir J 2004; 23(6): 932-946):

- Accessory respiratory muscle use
- Central cyanosis, new onset or worsening
- Paradoxical chest wall movements

52:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels.

PCO₂ values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

53:

In the setting of dyspnea, patients prefer sitting because they are unable to breathe effectively in the recumbent position.

54:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

55:

The administration of systemic corticosteroids in patients with a chronic obstructive pulmonary disease (COPD) exacerbation improves lung function, decreases hypoxemia, shortens recovery time, reduces hospital length of stay, and prevents early reoccurrence. Evidence demonstrates daily administration of 40 milligrams of prednisone in patients who have been hospitalized for more than 5 days is not inferior to a 14-day course in terms of early relapse. This shortened regimen is supported by current guidelines. Studies have shown low-dose oral steroids demonstrate equivalent outcomes to intravenous therapy in the treatment of COPD. Nebulized budesonide may be an alternative treatment for exacerbations with similar benefits to intravenous methylprednisolone in select patients, according to the GOLD guidelines (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

56:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

57:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

58:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

59:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation

2015).

60:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

61:

Right ventricular dysfunction confirmed by imaging demonstrates an increased risk for mortality. These patients should have intensive monitoring for potential hemodynamic decline and assessment for the need for possible intervention (e.g., thrombolysis, thrombectomy) (Kearon et al., Chest 2016, 149: 315-52).

62:**Expected progress after 48-72h**

- Anticoagulation progressing, if on warfarin, international normalized ratio (INR) increasing towards the therapeutic goal
- Pain resolved
- Vital signs stable or within acceptable limits
- Respiratory status improved
- No evidence of bleeding
- Increasing activity
- Adequate oral intake
- Patient or caregiver is able to manage anticoagulation regimen
- Preparing for discharge when INR is therapeutic or when anticoagulation teaching is completed

If no, consider:

- Adjusting dosage or changing anticoagulant agent and/or combination therapy, possible hematology or gastroenterology consultation
- Reassess pain medications
- Additional diagnostic testing: Imaging studies, (chest x-ray, computed tomography (CT), or magnetic resonance imaging (MRI)), to rule out additional pulmonary emboli or other respiratory disease processes, such as pneumonia
- Electrocardiogram (ECG) to rule out acute coronary syndrome (ACS)

63:

Instruction: This criteria point is intended for patients started on an unfractionated heparin (UFH) infusion on a previous episode day who require continued stay on UFH while receiving pretreatment prior to dabigatran or edoxaban initiation or bridging to warfarin.

64:

Instruction: This criterion applies to patients with actual, suspected, or a history of heparin-induced thrombocytopenia (HIT).

65:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

66:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

67:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack

of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

68:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

69:

Anticoagulants are among the most common drugs associated with medication errors. Before discharge, patients and/or caregivers should receive anticoagulant counseling tailored to their unique medication and delivered at an appropriate literacy level. The Joint Commission recommends that anticoagulation education cover the importance of follow-up monitoring, adherence, drug-food interactions, and the risk of adverse effects and drug interactions (Hawes, Pharmacy (Basel) 2018, 6: epub).

70:

The use of nonsteroidal anti-inflammatory medications or aspirin while on anticoagulant therapy has been associated with an increased risk for bleeding (den Exter et al., J Thromb Haemost 2022, 20: 1910-9).

71:

Mitigating risk factors (e.g., discontinuing hormone medication, smoking cessation) can aid in the prevention of future venous thromboembolism (Mount et al., Am Fam Physician 2022, 105: 377-85; Chopard et al., Jama 2020, 324: 1765-76).

72:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and

Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

73:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

74:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

75:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

76:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

77:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

78:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next

higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

79:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

80:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

81:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

82:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining

- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

83:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

84:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

85:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

86:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

87:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

88:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

89:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.